

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\text{SO}(2)$$

$$R^{-1} = R^T$$

$$\det R = 1$$

$$\text{SO}(2)$$

$$\det R = 1$$



$$z = e^{i\theta}$$

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$|z| = 1$$

$$R^{-1} = R^T$$

$$z = e^{i\theta}$$

$$e^{i\theta}$$

$$R(\theta)$$

$$|z| = 1$$

$$R_1 R_2 = R(\theta_1 + \theta_2)$$

$$R(\theta)$$

$$R_1 R_2 = R(\theta_1 + \theta_2)$$